

QUANTITATIVE RISK ASSESSMENT

Term	Definition
Single loss expectancy (SLE)	Total loss expected from a single incident
Annual rate of occurrence (ARO)	Number of times an incident is expected to occur in a year
Annual loss expectancy (ALE)	Expected loss for a year

$$\text{ALE} = \text{SLE} \times \text{ARO}$$

Safeguard value: Cost of a safeguard or control

SCENARIO 1:

Summit Ridge Outfitters is a Colorado-based outdoor clothing company with (20) employees. Each employee has a company-owned laptop valued at \$1,200. Over the past year, employees have lost or damaged (4) laptops.

With this information, calculate the following:

SLE	\$1200
ARO	4
ALE	\$4800

SAFEGUARD DECISION

Summit Ridge Outfitters is considering purchasing device protection insurance for each laptop. Use the ALE to determine the usefulness of this safeguard. For example, Summit Ridge Outfitters could purchase insurance for each laptop for \$85 per year.

Safeguard Value = \$85 x 20 devices = \$1,700

It is estimated that if insurance is purchased, the ARO will decrease to 1 (only 1 laptop lost/damaged per year).

DETERMINE THE EFFECTIVENESS OF THE SAFEGUARD:

Fill in the boxes below.

Current ALE	\$4800
ARO with control	1
ALE with control	\$1200
Saving with control (Current ALE – ALE with control)	\$3600
Safeguard value (cost of control):	\$1,700
Realized savings (Savings with control – safeguard value)	\$1900

SHOULD SUMMIT RIDGE OUTFITTERS BUY THE INSURANCE?

Why/why not should Summit Ridge Outfitters purchase the insurance? What would the positive/negative impact be? Explain your answer in 2-4 sentences.

Summit Ridge Outfitters should purchase the insurance. Saving \$1,900 every year would have the company recover more in reduced expected losses than the price of the insurance.

QUALITATIVE RISK ASSESSMENT

Term	Definition
Probability	The likelihood that a threat will exploit a vulnerability.
Impact	The negative result if a risk occurs.

CALCULATING RISK LEVEL: PROBABILITY X IMPACT

$$\text{Risk Level} = \text{Probability} \times \text{Impact}$$

Using these scales:

Probability	Impact
- Low= 10% - Medium = 50% - High = 100%	- Low = 10 - Medium 50 - High = 100

SCENARIO 2:

Summit Ridge Outfitters is concerned about protecting customer data and new product design files. Management identifies these primary risks:

- Unauthorized access by an external party
- Data theft or sabotage by an internal employee
- Hardware failure or device loss

After surveying key individuals, Summit Ridge Outfitters estimated probability and impact values as shown below.

Calculate the risk level for each risk.

Category	Probability	Impact	Risk Level
Unauthorized access by an external party	50	75	3750
Sabotage by an internal employee	25	100	2500
Hardware failures	75	25	1875

ANSWER THE FOLLOWING QUESTIONS:

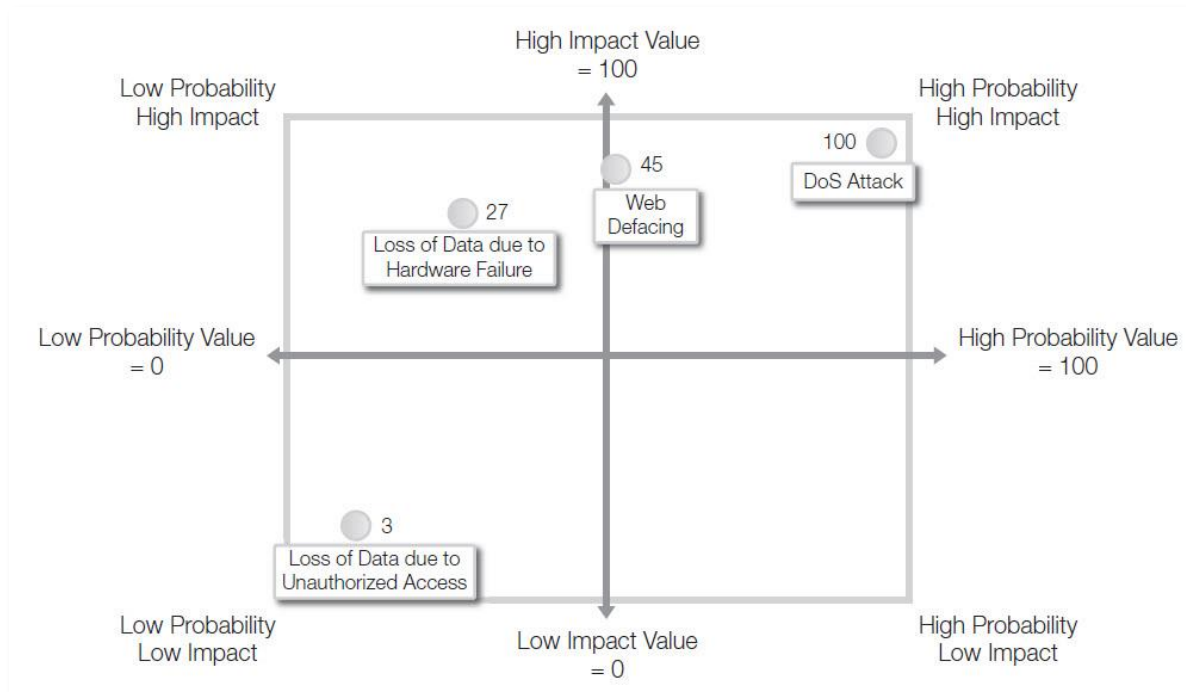
Which risk has the highest risk level?

Unauthorized access by an external party

Prioritize the risks from high to low:

Priority 1	Unauthorized access by an external party
Priority 2	Sabotage by an internal employee
Priority 3	Hardware failures

RISK MATRIX



Using the example matrix, complete the following risk matrix based on your data.

